

C loop3.c > ...

```
1  #include <stdio.h>
2
3  int main() {
4      float population = 100000;    // Initial population
5      float rate = 0.10;            // 10% increase per year
6      int year;
7
8      printf("Year\tPopulation\n");
9      printf("-----\n");
10
11     for (year = 1; year <= 10; year++) {
12         population = population + (population * rate); // Increase by 10%
13         printf("%d\t%.0f\n", year, population);
14     }
15
16     return 0;
17 }
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\ACER\Desktop\itscbe> ./a.exe

Year	Population

1	110000
2	121000
3	133100
4	146410
5	161051
6	177156
7	194872

0

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Q. The population of a town is 100000. The population has increased steadily at the rate of 10% per year for the last 10 yrs.

MAP to determine the population at the end of each year in the last decade

include <stdio.h>

int main() {

float population = 100000;

float rate = 0.10;

int year;

printf("Year\tPopulation\n");

printf("-----\n");

for (year = 1; year <= 10; year++) {

population = population + (population * rate);

printf("%1.0f\t%.0f\n", year, population);

}

return 0;

}

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itscbe

fin1.cfin2.cloop3.clo.cocloop4.cRelease Notes: 1.105.0fin3.cfin4.cfin5.cfin6.c

2

```
oc.c > main()
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, num = 1;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &n);
8
9      for (i = 0; i < n; i++) {
10         for (int space = 0; space < n - i; space++)
11             printf(" ");
12
13         num = 1;
14         for (j = 0; j <= i; j++) {
15             printf("%d ", num);
16             num = num * (i - j) / (j + 1);
17         }
18         printf("\n");
19     }
20
21     return 0;
22 }
23
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

PS C:\Users\ACER\Desktop\itscbe> gcc oc.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter number of rows: 3
1
1 1
1 2 1
PS C:\Users\ACER\Desktop\itscbe>

+ ... | x

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Ln 12, Col 1Spaces: 2UTF-8CRLF{}CGo LiveWin32

b)

```

      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1

```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, j, num = 1;
```

```
    printf("Enter number of rows:");
```

```
    scanf("%d", &n);
```

```
    for(i=0; i<n; i++) {
```

```
        for(int space=0; space<n-i; space++)
```

```
            printf(" ");
```

```
        num=1;
```

```
        for(j=0; j<=i; j++) {
```

```
            printf("%d", num);
```

```
            num = num * (i-j) / (j+1);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

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fin1.cfin2.cloop3.clo.cloop4.cRelease Notes: 1.105.0fin3.cfin4.cfin5.cfin6.cfin7.c

lo.c > ...

```
1 #include <stdio.h>
2
3 int main() {
4     int i, j, k = 1, space;
5     int rows = 3;
6
7     for (i = 1; i <= rows; i++) {
8         for (space = i; space < rows; space++) {
9             printf(" ");
10        }
11
12        for (j = 1; j <= i; j++) {
13            printf("%d ", k);
14            k++;
15        }
16
17        printf("\n");
18    }
19    return 0;
20 }
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

PS C:\Users\ACER\Desktop\itscbe> gcc lo.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
1
2 3
4 5 6
PS C:\Users\ACER\Desktop\itscbe>

+ ... | x

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Ln 2, Col 1Spaces: 2UTF-8CRLF{} CGo LiveWin32

Q. WAP to generate the following set of output.

a)

```
    1
   2 3
  4 5 6
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int i, j, k=1, space;
```

```
    int rows=3;
```

```
    for (i=1; i<=rows; i++) {
```

```
        for (space=i; space<rows; space++) {
```

```
            printf(" ");
```

```
        }
```

```
        for (j=1; j<=i; j++) {
```

```
            printf("%d", k);
```

```
            k++;
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

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fin1.cfin2.cloop1.cloop2.cfin3.cfin4.cfin5.cfin6.cfin7.cfin8.cfin9.c

loop2.c > ...

```
1  #include <stdio.h>
2
3  int main() {
4      int num, i;
5
6      // Input from user
7      printf("Enter a number to print its multiplication table: ");
8      scanf("%d", &num);
9
10     // Print the multiplication table
11     printf("\nMultiplication Table of %d:\n", num);
12     for (i = 1; i <= 10; i++) {
13         printf("%d * %d = %d\n", num, i, num * i);
14     }
15
16     return 0;
17 }
18
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Multiplication Table of 43:

43 * 1 = 43
43 * 2 = 86
43 * 3 = 129
43 * 4 = 172
43 * 5 = 215
43 * 6 = 258
43 * 7 = 301
43 * 8 = 344

+ ... | x

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Ln 18, Col 1 Spaces: 2 UTF-8 CRLF {} C Go Live Win32

```
printf("\nTotal negative no: %d", neg);  
printf("\nTotal zeroes: %d\n", zero);
```

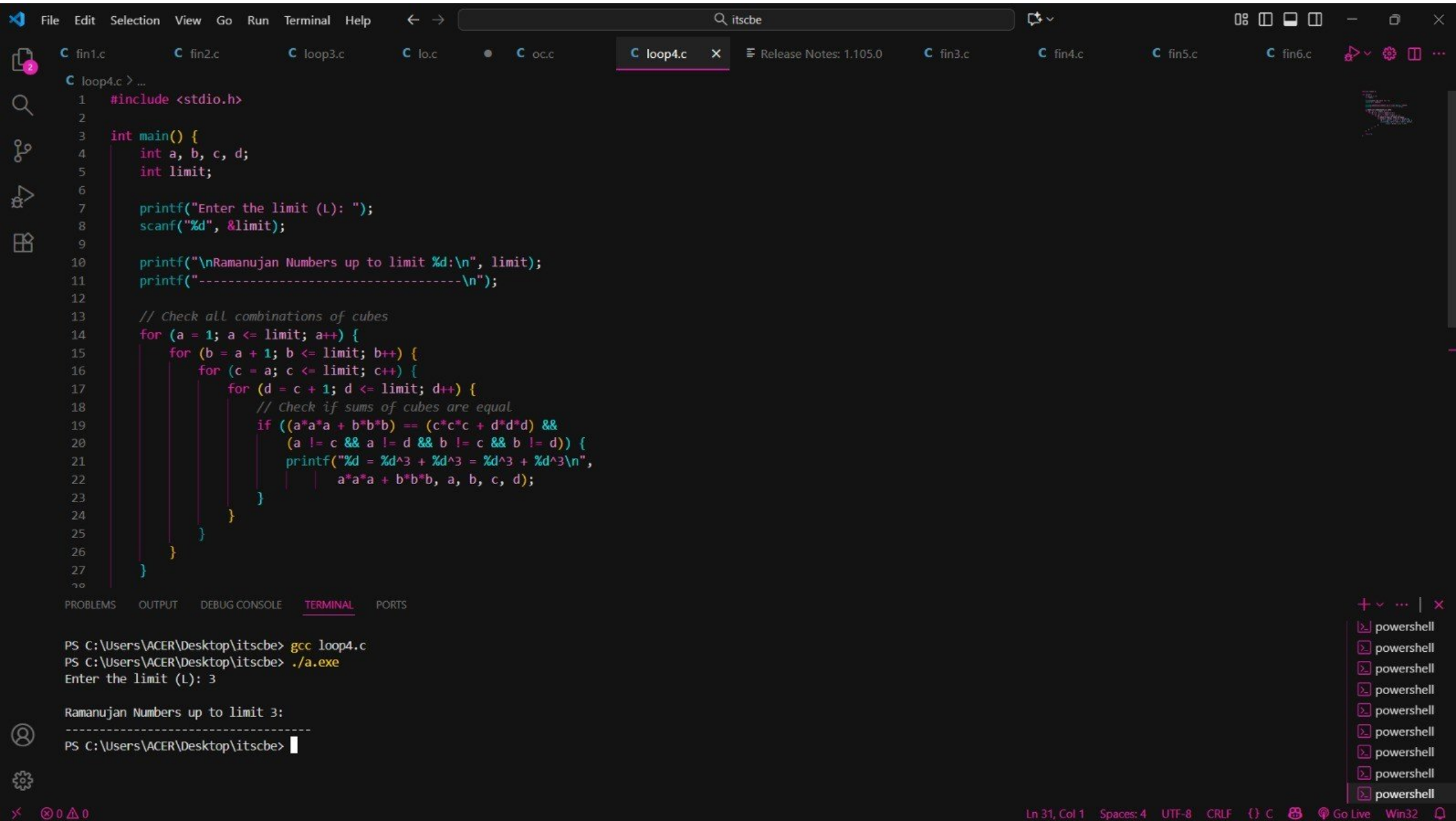
```
return 0;  
}
```

- Q. WAP to print the multiplication table of the number entered by the user.
It should be in the correct formatting.
Num * 1 = Num

→ #include <stdio.h>

```
int main() {  
    int num, i;  
    printf("Enter a number to print its multiplication table.");  
    scanf("%d", &num);
```

```
    printf("\nMultiplication Table of %d: \n", num);  
    for (i = 1; i <= 10; i++) {  
        printf("%d * %d = %d\n", num, i, num * i);  
    }  
    return 0;  
}
```

Q. Ramanujan Number is the smallest number that can be expressed as the sum of ~~the~~ two cubes in 2 diff ways. WAP to print all such numbers upto a reasonable limit.

```
→ #include <stdio.h>
int main() {
    int a, b, c, d;
    int limit;
```

```
    printf("Enter the limit (1):");
    scanf("%d", &limit);
```

```
    printf("\nRamanujan Number up to limit %d : n", limit);
    printf("- - - - - \n");
```

```
    for(a=1; a <= limit; a++) {
        for(b=a+1; b <= limit; b++) {
            for(c=a; c <= limit; c++) {
                for(d=c+1; d <= limit; d++) {
```

```
                    if (a*a*a + b*b*b == (c*c*c + d*d*d) &&
                        (a != c && a != d && b != c && b != d)) {
```

```
                        printf("%d = %d^3 + %d^3 = %d^3 + %d^3 \n", a*a*a
                            + b*b*b, a, b, c, d);
```

```
                    } } } } }
                }
            }
        }
    }
    return 0;
}
```

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fin1.cfin2.cloop1.cfin3.cfin4.cfin5.cfin6.cfin7.cfin8.cfin9.c

loop1.c > ...

```
1  #include <stdio.h>
2
3  int main() {
4      int num;
5      int pos = 0, neg = 0, zero = 0;
6      char choice;
7
8      do {
9          // Input a number
10         printf("Enter a number: ");
11         scanf("%d", &num);
12
13         // Check if positive, negative, or zero
14         if (num > 0)
15             pos++;
16         else if (num < 0)
17             neg++;
18         else
19             zero++;
20
21         // Ask user if they want to continue
22         printf("Do you want to enter another number? (y/n): ");
23         scanf(" %c", &choice); // note the space before %c to consume newline
24
25     } while (choice == 'y' || choice == 'Y');
26
27     // Display the results
28     printf("\nTotal Positive numbers: %d\n", pos);
29 }
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Do you want to enter another number? (y/n): y
Enter a number: 2
Do you want to enter another number? (y/n): y
Enter a number: 0
Do you want to enter another number? (y/n): n

Total Positive numbers: 2
Total Negative numbers: 0
Total Zeroes: 1
PS C:\Users\ACER\Desktop\itscbe>

+ ... | x

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Ln 34, Col 1 Spaces: 2 UTF-8 CRLF {} C Go Live Win32

loops

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Q WAP to enter numbers till the user wants. At the end, it should display the count of positive, negative and zeroes entered.

```
int main() {  
    char choice;  
  
    do {  
        printf("Enter a number:");  
        scanf("%d", &num);  
  
        if (num > 0)  
            pos++;  
        else if (num < 0)  
            neg++;  
        else  
            zero++;  
    } while (choice == 'y' || choice == 'Y');
```

```
printf("Do you want to enter number? (y/n):");  
scanf("%c", &choice);
```

```
while (choice == 'y' || choice == 'Y');
```

```
printf("\nTotal positive numbers: %d", pos);
```

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